

# Relation Algebra

## §1. Notation

Traditionally Adjunction  $F \dashv G$  is defined as  $F(x) \leq y$  iff  $x \leq G(y)$ , and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define  $X : * \mapsto x$ ,  $Y : * \mapsto y$  and use diagram order, application can be replaced by composition:  $XF \leq Y$  iff  $X \leq YG$

(1a)

- write  $F$  as  $/$ ,  $G$  as  $\backslash$
- render  $a \leq b$  as  $\boxed{\frac{a}{b}}$
- **rewrite**  $X \leq XF$  as  $1 \leq F$

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj



now we have  $\bigwedge = /$  and  $\bigvee = \backslash$ , and you just discovered string diagrams.

## §1.1. Adjunctions

(1.1a)

| $F \dashv G$                                           | $F$ 's type                                                           | monad $FG$                            | $1 \sqsubseteq FG$                                                         | $GF \sqsubseteq 1$                                                         | $F$ preserves $u$                                                 | $G$ preserves $n$                                                         | $FGF=F$                                                       | $GFG=G$                                                         |
|--------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------|
| $\triangleleft \dashv \triangleleft$                   | $A \rightarrow A \otimes A$                                           | $\triangleleft \triangleleft = 1$     | $1 \sqsubseteq \triangleleft \triangleleft$                                | $\triangleleft \triangleleft \sqsubseteq 1$                                | $(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$ | $(R \cap S) \triangleright = R \triangleright \cap S \triangleright$      | $\triangleleft \triangleleft \triangleleft = \triangleleft$   | $\triangleright \triangleright \triangleright = \triangleright$ |
| $\multimap \dashv \multimap$                           | $A \rightarrow I$                                                     | $\multimap \multimap = T$             | $1 \sqsubseteq \multimap \multimap$                                        | $\multimap \multimap \sqsubseteq 1$                                        | $(R \cup S) \multimap = R \multimap \cup S \multimap$             | $(R \cap S) \multimap = R \multimap \cap S \multimap$                     | $\multimap \multimap \multimap = \multimap$                   | $\multimap \multimap \multimap = \multimap$                     |
| $\circ \dashv \circ$                                   | $(A \rightarrow B) \rightarrow (B \rightarrow A)$                     | $R^\circ \circ = R$                   | $R^\circ \circ = R$                                                        | $R^\circ \circ = R$                                                        | $(R \cup S)^\circ = S^\circ \cup R^\circ$                         | $(R \cap S)^\circ = R^\circ \cap S^\circ$                                 | $R^\circ \circ = R$                                           | $R^\circ \circ = R$                                             |
| $\circ \dashv \triangleleft \dashv \multimap$          | $(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$ | $1$                                   | $(\circ \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$ | $(1 \otimes \circ \triangleleft) (\triangleright \multimap \otimes 1) = 1$ | $=$                                                               | $=$                                                                       | $=$                                                           | $=$                                                             |
| $\triangleright \dashv \multimap \dashv \triangleleft$ | $(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$ | $1$                                   | $(1 \otimes \circ \triangleleft) (\triangleright \multimap \otimes 1) = 1$ | $(\circ \triangleleft \otimes 1) (1 \otimes \triangleright \multimap) = 1$ | $=$                                                               | $=$                                                                       | $=$                                                           | $=$                                                             |
| $\Delta \dashv \cap$                                   | $(A \rightarrow B) \rightarrow (A \rightarrow B)^2$                   | $R \cap R = R$                        | $R \cap R = R$                                                             | $R \cap S \sqsubseteq R$                                                   | $\Delta(R \cup S) = \Delta R \cup \Delta S$                       | $(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$                 | $R \cap R = R$                                                | $R \cap R = R$                                                  |
| $u \dashv \Delta$                                      | $(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$                   | $(R, S) \mapsto (R \cup S, R \cap S)$ | $R \sqsubseteq R \cup S$                                                   | $R \cup R = R$                                                             | $(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$         | $\Delta(R \cap S) = \Delta R \cap \Delta S$                               | $R \cup R = R$                                                | $R \cup R = R$                                                  |
| $\emptyset \dashv !$                                   | $\{*\} \rightarrow (A \rightarrow B)$                                 |                                       |                                                                            | $\emptyset \sqsubseteq R$                                                  |                                                                   |                                                                           |                                                               |                                                                 |
| $\cdot S \dashv / S$                                   | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $S/S$                                 | $R \sqsubseteq (RS)/S$                                                     | $(T/S) S \sqsubseteq T$                                                    | $(S \cup T)R = SR \cup TR$                                        | $(R \cap R')/S = (R/S) \cap (R'/S)$                                       | $((RS)/S)S = RS$                                              | $((T/S)S)/S = T/S$                                              |
| $S \cdot \dashv S \setminus$                           | $(B \rightarrow C) \rightarrow (A \rightarrow C)$                     | $S \setminus S$                       | $R \sqsubseteq S \setminus (SR)$                                           | $S(S \setminus T) \sqsubseteq T$                                           | $R(S \cup T) = RS \cup RT$                                        | $S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$         | $S(S \setminus (SR)) = SR$                                    | $S \setminus (S(S \setminus T)) = S \setminus T$                |
| $R \cap \dashv R \Rightarrow$                          | $(A \rightarrow B) \rightarrow (A \rightarrow B)$                     | $X \mapsto R \Rightarrow (X \cap R)$  | $X \sqsubseteq R \Rightarrow (X \cap R)$                                   | $R \cap (R \Rightarrow Y) \sqsubseteq Y$                                   | $R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$                  | $R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$       | $R \cap (R \Rightarrow (X \cap R)) = X \cap R$                | $R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$    |
| $\mathcal{D} \dashv \cdot T$                           | $(A \rightarrow B) \rightarrow \text{Cor } A$                         | $R \mapsto (\mathcal{D}R)T$           | $R \sqsubseteq (\mathcal{D}R)T$                                            | $\mathcal{D}(AT) \sqsubseteq A$                                            | $\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$          | $(A \cap B)T = AT \cap BT$                                                | $\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$                 | $(\mathcal{D}(AT))T = AT$                                       |
| $\mathcal{R} \dashv T \cdot$                           | $(A \rightarrow B) \rightarrow \text{Cor } B$                         | $R \mapsto T(\mathcal{R}R)$           | $R \sqsubseteq T(\mathcal{R}R)$                                            | $\mathcal{R}(TA) \sqsubseteq A$                                            | $\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$          | $T(A \cap B) = TA \cap TB$                                                | $\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$                 | $T(\mathcal{R}(TA)) = TA$                                       |
| $\cdot f \dashv \cdot f^\circ$                         | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $ff^\circ$                            | $\text{Map}(f) \Rightarrow 1 \sqsubseteq ff^\circ$                         | $f^\circ f \sqsubseteq 1$                                                  | $(S \cup T)R = SR \cup TR$                                        | $\text{Simple}(g) \Rightarrow (X \cap Y)g^\circ = Xg^\circ \cap Yg^\circ$ | $ff^\circ f = f$                                              | $f^\circ ff^\circ = f^\circ$                                    |
| $f^\circ \dashv \cdot f \cdot$                         | $(A \rightarrow C) \rightarrow (B \rightarrow C)$                     | $ff^\circ$                            | $\text{Map}(f) \Rightarrow 1 \sqsubseteq ff^\circ$                         | $f^\circ f \sqsubseteq 1$                                                  | $R(S \cup T) = RS \cup RT$                                        | $f(X \cap Y) = fX \cap fY$                                                | $ff^\circ f = f$                                              | $f^\circ ff^\circ = f^\circ$                                    |
| $i \dashv E$                                           | $\text{Map} \dashv \text{Rel}$                                        | $E$                                   | $\frac{1}{\exists} : A \rightarrow EA$                                     | $\exists : EB \rightarrow B$                                               | —                                                                 | —                                                                         | $\frac{\exists}{\exists} = 1$                                 | $\frac{R}{\exists} \exists = R$                                 |
| $\cdot \exists \dashv \frac{\exists}{\exists}$         | $\text{Map}(A, EB) \rightarrow (A \rightarrow B)$                     | $1$                                   | $\frac{f\exists}{\exists} = f$                                             | $\frac{R}{\exists} \exists = R$                                            | —                                                                 | —                                                                         | $\frac{\frac{\exists}{\exists}}{\exists} = f\exists$          | $\frac{\frac{R}{\exists}}{\exists} = \frac{R}{\exists}$         |
| $\frac{\exists}{\exists} \dashv \cdot \exists$         | $(A \rightarrow B) \rightarrow \text{Map}(A, EB)$                     | $1$                                   | $\frac{R}{\exists} \exists = R$                                            | $\frac{f\exists}{\exists} = f$                                             | —                                                                 | —                                                                         | $\frac{\frac{\exists}{\exists}}{\exists} = \frac{R}{\exists}$ | $\frac{\frac{f\exists}{\exists}}{\exists} = f\exists$           |

## §1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
 \begin{array}{ccccc}
 \boxed{\begin{array}{c} X \\ f(R/T) \end{array}} & \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} & \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
 \text{f a map} & & & & \\
 \xLeftrightarrow{\text{associativity}} & \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} & \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} & \boxed{\begin{array}{c} XT \\ fR \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
 \xRightarrow{\text{indirect equality}} & & & & & \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
 \end{array}$$

$$\text{Map}(f) \Rightarrow f(R/S) = fR/S$$

(1.2b)

|                       | $\cdot T$                                    | $T \cdot$                                                 | $\cdot g$                                         | $g^\circ \cdot$                              | $Tn$                                                      | $^\circ$                                                                                                                            | $\circ \triangleleft$                                                                                                               | $\Delta$                                                       |
|-----------------------|----------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|----------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| $\cdot S$             | $R/S_1 \ S_2=R/S_2/S_1$                      | $S \setminus (R/W) = (S \setminus R)/W$                   | $R/(Sg) = (Rg^\circ)/S$                           | $\text{Map}(f) \implies f(R/S) = fR/S$       | —                                                         | $(Y/S)^\circ = S^\circ \setminus (Y^\circ)$                                                                                         | $(RS)^\wedge = R^\wedge (S \otimes 1)$                                                                                              | $(RnR')/S = (R/S) \cap (R'/S)$                                 |
| $S \cdot$             | $S \setminus (R/W) = (S \setminus R)/W$      | $S_1 \ S_2 \setminus R = S_2 \setminus (S_1 \setminus R)$ | $S \setminus (Rg^\circ) = (S \setminus R)g^\circ$ | $(g^\circ S) \setminus R = S \setminus (gR)$ | —                                                         | $(S \setminus Y)^\circ = Y^\circ / S^\circ$                                                                                         | $((S \otimes 1)R)^\wedge = SR^\wedge$                                                                                               | $S \setminus (RnR') = (S \setminus R) \cap (S \setminus R')$   |
| $\cdot f$             | $\text{Map}(f) \implies R/fS = (R/S)f^\circ$ | $T \setminus (Rf^\circ) = (T \setminus R)f^\circ$         | $(RS)^\circ = S^\circ R^\circ$                    |                                              |                                                           | $(RS)^\circ = S^\circ R^\circ$                                                                                                      |                                                                                                                                     | $\text{Simple}(g) \implies (XnY)g^\circ = Xg^\circ nYg^\circ$  |
| $f^\circ \cdot$       | $\text{Map}(f) \implies f(R/S) = fR/S$       | $(Tf^\circ) \setminus R = f(T \setminus R)$               | —                                                 | $(RS)^\circ = S^\circ R^\circ$               |                                                           | $(RS)^\circ = S^\circ R^\circ$                                                                                                      |                                                                                                                                     | $f(T_1 n T_2) = fT_1 n fT_2$                                   |
| $Rn$                  | —                                            | —                                                         | —                                                 | —                                            | $R \Rightarrow S \Rightarrow T = RnS \Rightarrow T$       | $(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$                                                                           | —                                                                                                                                   | $R \Rightarrow SnT = (R \Rightarrow S) \cap (R \Rightarrow T)$ |
| $^\circ$              | $(Y/T)^\circ = T^\circ \setminus (Y^\circ)$  | $(T \setminus Y)^\circ = Y^\circ / T^\circ$               | $(RS)^\circ = S^\circ R^\circ$                    | $(RS)^\circ = S^\circ R^\circ$               | $(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$ | $R^\circ \circ = R$                                                                                                                 | $(R^\wedge)^\circ = (R^\circ \otimes 1)$<br>$(1 \otimes \triangleright \dashv \circ)$<br>both bends: the $^\circ$ section's display | $(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$                    |
| $\circ \triangleleft$ | $(RT)^\wedge = R^\wedge (T \otimes 1)$       | $((T \otimes 1)R)^\wedge = TR^\wedge$                     | —                                                 | —                                            | —                                                         | $(R^\wedge)^\circ = (R^\circ \otimes 1)$<br>$(1 \otimes \triangleright \dashv \circ)$<br>both bends: the $^\circ$ section's display | —                                                                                                                                   | —                                                              |
| $u$                   | $(S \cup T)R = SR \cup TR$                   | $R(S \cup T) = RS \cup RT$                                | $(S \cup T)R = SR \cup TR$                        | $R(S \cup T) = RS \cup RT$                   | $Rn(S \cup T) = RnS \cup RnT$                             | $(R \cup S)^\circ = S^\circ \cup R^\circ$                                                                                           |                                                                                                                                     |                                                                |
| $\emptyset$           | $\emptyset R = \emptyset$                    | $R \emptyset = \emptyset$                                 | $\emptyset R = \emptyset$                         | $R \emptyset = \emptyset$                    | $Rn \emptyset = \emptyset$                                | $\emptyset^\circ = \emptyset$                                                                                                       |                                                                                                                                     |                                                                |

### §1.3. Adjoint triples

Beyond  $u \dashv \Delta \dashv n$ , which §Section 1.1 already carries as two rows, the table holds one adjoint triple with content:  $\exists\_f \dashv f^* \dashv \forall\_f$  along a map  $f$ , once on each side of the composite.  $f : A \rightarrow B$  throughout this subsection.

(1.3a)

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot \dashv f \cdot \dashv f \setminus \end{aligned}$$

The first two links of each chain are the map shunting rules, and the third is §Section 1.1's division row read at a chosen divisor:  $\cdot S \dashv /S$  at  $S := f^\circ$ , and  $S \cdot \dashv S \setminus$  at  $S := f$ .

The third link is not the first over again. For a map  $f$ ,  $/f$  collapses to  $\cdot f^\circ$ ; being a map is exactly what that collapse spends  $\text{Map}(f) \implies R/fS = (R/S)f^\circ$  —  $f^\circ$  is not a map, so  $/f^\circ$  does not collapse in turn, and the three operators stay distinct. On  $\text{Rel}(\mathbb{I}, A) = \text{EA}$  they are the image triple:

(1.3b)

| operator        | acts by                                   | name                       |
|-----------------|-------------------------------------------|----------------------------|
| $\cdot f$       | $S \mapsto f[S]$                          | direct image, $\exists\_f$ |
| $\cdot f^\circ$ | $T \mapsto f^{-1}[T]$                     | inverse image, $f^*$       |
| $/f^\circ$      | $S \mapsto \{b : f^{-1}(b) \subseteq S\}$ | $\forall\_f$               |

§Section 1.1's  $\mathcal{D} \dashv \cdot T$  is this same chain along the projection  $A \otimes B \rightarrow A$ , read through  $\text{Rel}(A, B) = E(A \otimes B)$ :  $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$ ,  $AT$  is the relation that ignores  $b$  altogether, and the third link is  $R \mapsto 1 \cap R / T = \{(a, a) : \forall b. aRb\}$ .

(1.3c)

$$\mathcal{D} \dashv \cdot T \dashv 1 \cap \cdot / T$$

Three is where it stops, and one map breaks both ends. Take  $A = \{a_1, a_2\}$  and  $B = \{b\}$ :  $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$  while  $f[\{a_1\} \cap \{a_2\}] = \emptyset$ , so  $\cdot f$  does not preserve meets and has no left adjoint, and the same two subsets give  $\forall\_f(\{a_1\} \cup \{a_2\}) = \{b\}$  against  $\forall\_f\{a_1\} \cup \forall\_f\{a_2\} = \emptyset$ , so  $/f^\circ$  does not preserve joins and has no right adjoint. A monic  $f$  restores the binary case and no more —  $Tf$  still reaches only  $\text{im } f$ , and  $\forall\_f \emptyset = B \setminus \text{im } f$  is still not  $\emptyset$ . The chain extends only when  $f^\circ$  is a map as well, and then  $/f^\circ = \cdot f^\circ = \cdot f$  and it repeats forever. For an  $S$  with neither  $S$  nor  $S^\circ$  a map there is no triple at all:  $\cdot S \dashv /S$  is two links and stops.

(1.2b)

The rows that chain forever say nothing by it:  $\circ$  is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and  $\circ \dashv \circ \dashv \circ$  has no end, and §Section 1.1’s other row of that kind,  $\circ \triangleleft \dashv \triangleright \circ$ , goes the same way.